

Yuvraj Kashyap

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Software Engineer

EDUCATION

University of Texas at Dallas

Expected May 2027

B.S. in Computer Science, Minor in Finance, Certificate in Entrepreneurship & Innovation

GPA: 3.76/4.00 | Dean's List | NCAA Division II Men's Tennis

Relevant Coursework: Advanced Algorithms, Data Structures & Algorithms, Database Systems, Systems Programming, Software Engineering, Programming Languages, Machine Learning, Artificial Intelligence, Computer Vision, Linear Algebra, Probability & Statistics

EXPERIENCE

Software Engineering Intern, Medceptor

Apr 2026 – Present

IDK Studios

- Shipped production full-stack features for Medceptor, an AI-driven medical education platform, using Next.js, React, TypeScript, Django, Supabase/Postgres, Vercel, Doppler, and GitHub in a small startup team.
- Delivered 4+ PRs building RN question-bank and exam-mode workflows supporting 100+ generated questions, multiple formats, track/level filtering, import/validation pipelines, scoring, review flows, and release QA for CNA/LPN/RN expansion.

Undergraduate Researcher, UAV / Smart City Systems

Oct 2025 – Present

University of Texas at Dallas

- Built a geometry-driven UAV swarm simulation framework over a 4 km × 4 km OpenStreetMap/Blender urban environment to evaluate multi-agent path planning, visibility, and coverage under realistic city constraints.
- Modeled occlusions, non-convex obstacles, sensing constraints, and learning-guided coordination heuristics to study UAV behavior beyond simplified grid or voxel-based environments.

Electrical Engineer, NOVA Autonomous Driving Research Program

Sep 2023 – Jun 2025

University of Texas at Dallas

- Built Python preprocessing pipelines for 3D sensor data, including voxel-grid point cloud downsampling that reduced scan size from 370,277 to 20,528 points while preserving geometric structure.
- Debugged power, sensor, and I/O reliability issues across autonomous vehicle subsystems, translating hardware constraints into software and systems-level fixes for perception reliability.

Vice President of Finance & Senior Consultant, Consult Your Community

Jan 2026 – Present

University of Texas at Dallas

- Led consulting teams for Glydr and Get in the Path; built Glydr's web-based gaming hub prototype and drove product operations, pricing/financial strategy, go-to-market planning, marketing, nonprofit launch operations, and final client deliverables.

PROJECTS

Aletheia – Hybrid Retrieval, Reranking & Search Evaluation Platform

Python, FastAPI, PyTorch, PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Docker, Next.js

- Engineered a search and ranking platform comparing BM25, dense retrieval, hybrid Reciprocal Rank Fusion, and cross-encoder reranking with query tracing, experiment comparison, and reproducible evaluation workflows.
- Benchmarked 5,183 BEIR SciFact documents across 300 queries and 339 relevance judgments; hybrid RRF achieved 0.8429 Recall@10 with zero failed queries.

Atlas – Fault-Tolerant Distributed Crawl, Index & Search Platform

Python, FastAPI, PostgreSQL, Redis/RQ, OpenSearch, Docker, AWS, Terraform, Next.js

- Built a distributed crawl and search platform with PostgreSQL-authoritative scheduling, Redis-backed workers, robots.txt/politeness enforcement, content extraction, deduplication, and OpenSearch retrieval.
- Passed a deterministic 10,000-page fault benchmark covering worker loss, complete Redis data loss, retries, duplicate delivery, redirects, malformed content, and transient failures.

Beyond Chat – Artifact-Centric AI Workspace with Finance Agent

React, TypeScript, FastAPI, Supabase, OpenRouter, Exa, Stripe, LangChain, Vercel Sandbox, Pytest, Vitest

- Built a full-stack AI workspace for chat, writing, research, image, data, and finance workflows with workspace-scoped RLS, Stripe billing, artifact storage/search, streaming responses, and Dexter finance-agent execution through Vercel Sandbox.

Sticky – Production Task Platform with Durable Workflows

TypeScript, Next.js, Hono, Supabase Auth/Postgres, TanStack Query, Zod, Playwright, Vitest, Vercel

- Built a full-stack task platform with a versioned Hono command API, recurring work, reminders, Realtime cache invalidation, search, calendar planning, PWA support, and desktop/mobile workflows.
- Implemented idempotent mutations, transactional outbox delivery, encrypted server-only credentials, owner-scoped RLS, explicit database grants, shared Zod contracts, and end-to-end security and persistence tests.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

Backend & APIs: FastAPI, Django, Hono, Node.js, REST APIs, Server Actions, SQLAlchemy, Alembic, Pydantic, Zod

Data, Search & Infrastructure: PostgreSQL, Redis/RQ, OpenSearch, Qdrant, Supabase Auth/RLS, Docker, Docker Compose, Git, GitHub Actions, Vercel, Doppler, Linux/Unix, CI/CD

AI/ML Systems: BM25, vector search, embeddings, cross-encoder reranking, retrieval evaluation, OpenRouter, Exa, LangChain, PyTorch, NumPy, Pandas, OpenCV

Frontend & Integrations: React, Next.js, TanStack Query, Tailwind CSS, Vite, Framer Motion, Recharts, Stripe, Resend, QStash